

P8X Programmer's Guide

Instruction set rev 1 - generated from the microcode source (genucode.py); opcode values, byte and cycle counts are extracted from the same tables that build the EPROM images, so this document cannot drift from the hardware.

Programming model

A, B - 8-bit ALU operand registers; results land in A. **P0-P3** - four 16-bit pointer registers; the address bus is always driven by one of them. **P0** is the program counter. **P3** is the stack pointer (empty-descending: push writes then decrements; initialise it early - see note 3). P1 and P2 are general pointers used by the (Pn) addressing modes. **FLAGS** - C, Z, N, V, latched only by instructions marked in the Flags column.

Memory map (single-sourced in generators/gen_memmap.py): \$0000-\$17FF ROM (6K) | \$1800-\$FEFF RAM (\$1800-\$1FFF scratch island) | \$FF00-\$FFFF I/O: \$FF00 switches (r), \$FF02 LEDs (w), \$FF04 ACIA status (bit0 RX ready, bit1 TX ready), \$FF05 ACIA data.

Reset: P0 is forced to \$0000; execution begins there. All other registers (including P3) are undefined on real hardware.

Instruction set

Op	Mnemonic	Bytes	Cycles	Flags	Description
System					
\$00	NOP	1	2	-	No operation.
\$01	HLT	1	2	-	Halt the clock. Resume only by reset (or emulator exit).
\$72	CLC	1	2	C	C := 0.
\$73	SEC	1	2	C	C := 1.
Interrupts (rev C)					
\$02	EI	1	2	-	Enable maskable interrupts (IE := 1).
\$03	DI	1	2	-	Disable maskable interrupts (IE := 0).
\$04	RTI	1	8	-	Return from interrupt: pop flags then PC; re-enables IE.
\$08	IRQ	1	10	-	Software interrupt: push PC+flags, vector to \$0808. Also the opcode the hardware forcing buffer injects on a maskable IRQ.
Load / store					
\$10	LDA #imm	2	2	-	A := immediate byte.
\$11	LDB #imm	2	2	-	B := immediate byte.
\$12	LDA addr	3	6	Z N	A := byte at addr (absolute).
\$13	LDB addr	3	6	Z N	B := byte at addr (absolute).
\$14	STA addr	3	6	-	Byte at addr := A (absolute).
\$15	LDA (P1)+	1	2	-	A := memory at P1, then P1 := P1 + 1.
\$16	LDA (P2)+	1	2	-	A := memory at P2, then P2 := P2 + 1.
\$17	LDA (P3)+	1	2	-	A := memory at P3, then P3 := P3 + 1.
\$19	STA (P1)+	1	2	-	Memory at P1 := A, then P1 := P1 + 1.
\$1A	STA (P2)+	1	2	-	Memory at P2 := A, then P2 := P2 + 1.
\$1B	STA (P3)+	1	2	-	Memory at P3 := A, then P3 := P3 + 1.
\$1D	STA (P1)	1	2	-	Memory at P1 := A.
\$1E	STA (P2)	1	2	-	Memory at P2 := A.
\$1F	STA (P3)	1	2	-	Memory at P3 := A.
\$51	LDA (P1)	1	2	Z N	A := memory at P1 (P1 unchanged).
\$52	LDA (P2)	1	2	Z N	A := memory at P2 (P2 unchanged).
\$53	LDA (P3)	1	2	Z N	A := memory at P3 (P3 unchanged).

Op	Mnemonic	Bytes	Cycles	Flags	Description
\$88	LDA (P1+d)	2	7	C Z N	A := byte at P1 + d (d unsigned 0..255). C/V from the address add, then Z/N from A. P1 unchanged.
\$89	LDA (P2+d)	2	7	C Z N	A := byte at P2 + d (d unsigned 0..255). C/V from the address add, then Z/N from A. P2 unchanged.
\$8A	LDA (P3+d)	2	7	C Z N	A := byte at P3 + d (d unsigned 0..255). C/V from the address add, then Z/N from A. P3 unchanged.
\$8C	STA (P1+d)	2	9	C Z N	Byte at P1 + d := A. A preserved; flags clobbered by the address add.
\$8D	STA (P2+d)	2	9	C Z N	Byte at P2 + d := A. A preserved; flags clobbered by the address add.
\$8E	STA (P3+d)	2	9	C Z N	Byte at P3 + d := A. A preserved; flags clobbered by the address add.
ALU (operands A,B; result to A unless noted)					
\$20	ADD	1	2	C Z N	A := A + B.
\$21	SUB	1	2	C Z N	A := A - B.
\$22	AND	1	2	C Z N	A := A AND B.
\$23	OR	1	2	C Z N	A := A OR B.
\$24	XOR	1	2	C Z N	A := A XOR B.
\$25	CMP	1	2	C Z N	Flags from A - B; A unchanged.
\$26	INC	1	2	C Z N	A := A + 1. (B not used.)
\$27	DEC	1	2	C Z N	A := A - 1. (B not used.)
\$28	SHL	1	2	C Z N	A := A << 1, 0 into bit 0. (1)
\$29	SHR	1	2	C Z N	A := A >> 1, 0 into bit 7. (1)
\$2A	ROL	1	2	C Z N	Rotate A left through carry.
\$2B	ROR	1	2	C Z N	Rotate A right through carry.
ALU with T (rev C; 2nd operand = T register via B-mux; B preserved)					
\$80	ADDT	1	2	C Z N	A := A + T. (B preserved.)
\$81	SUBT	1	2	C Z N	A := A - T. (B preserved.)
\$82	ANDT	1	2	C Z N	A := A AND T. (B preserved.)
\$83	ORT	1	2	C Z N	A := A OR T. (B preserved.)
\$84	XORT	1	2	C Z N	A := A XOR T. (B preserved.)
\$85	CMPT	1	2	C Z N	Flags from A - T; A and B unchanged.
\$86	LDT #imm	2	2	-	T := immediate byte. (No flags.)
\$87	LDT addr	3	6	-	T := byte at addr (absolute). (No flags.)
Pointer registers					
\$31	LPL1 #imm	2	3	-	Low byte of P1 := immediate.
\$32	LPL2 #imm	2	3	-	Low byte of P2 := immediate.
\$33	LPL3 #imm	2	3	-	Low byte of P3 := immediate.
\$35	LPH1 #imm	2	3	-	High byte of P1 := immediate.
\$36	LPH2 #imm	2	3	-	High byte of P2 := immediate.
\$37	LPH3 #imm	2	3	-	High byte of P3 := immediate.
\$54	INP1	1	2	-	P1 := P1 + 1.
\$55	INP2	1	2	-	P2 := P2 + 1.
\$56	INP3	1	2	-	P3 := P3 + 1.
\$58	DEP1	1	2	-	P1 := P1 - 1.
\$59	DEP2	1	2	-	P2 := P2 - 1.

Op	Mnemonic	Bytes	Cycles	Flags	Description
\$5A	DEP3	1	2	-	P3 := P3 - 1.
\$5E	TAP1L	1	2	-	Low byte of P1 := A.
\$5F	TAP1H	1	2	-	High byte of P1 := A.
\$60	TAP2L	1	2	-	Low byte of P2 := A.
\$61	TAP2H	1	2	-	High byte of P2 := A.
\$62	TAP3L	1	2	-	Low byte of P3 := A.
\$63	TAP3H	1	2	-	High byte of P3 := A.
\$68	TPA1L	1	2	Z N	A := low byte of P1.
\$69	TPA1H	1	2	Z N	A := high byte of P1.
\$6A	TPA2L	1	2	Z N	A := low byte of P2.
\$6B	TPA2H	1	2	Z N	A := high byte of P2.
\$6C	TPA3L	1	2	Z N	A := low byte of P3.
\$6D	TPA3H	1	2	Z N	A := high byte of P3.
Stack					
\$70	PHA	1	2	-	Push A onto the P3 stack.
\$71	PLA	1	3	Z N	Pop A from the P3 stack.
16-bit memory ops (rev D; compiler space savers). PHW/PLW/LPW pure-microcode; MOVW adds the PT2 scratch pointer					
\$74	PHW addr	3	9	-	Push the 16-bit word at addr onto the P3 stack: high byte first, then low, so the word lies LITTLE-ENDIAN at P3+1..P3+2 (readable with LDW a,(P3+d); same layout as a JSR return address).
\$75	PLW addr	3	10	-	Pop a 16-bit word from the P3 stack into addr (low then high).
\$76	LPW1 addr	3	9	-	P1 (16-bit) := the word at addr.
\$77	LPW2 addr	3	9	-	P2 (16-bit) := the word at addr.
\$78	MOVW dst,src	5	13	-	16-bit memory->memory move: the word at src -> dst (via the PT/PT2 scratch pointers).
\$79	LPW3 addr	3	9	-	P3 (16-bit) := the word at addr -- restore a saved stack pointer.
\$BD	PHW (P1+d)	2	10	C Z N	Push the 16-bit word at P1 + d onto the P3 stack, high byte first (little-endian at the new P3+1) -- the C compiler's argument push straight from a frame slot. A! (the address add); flags from that add.
\$BE	PHW (P2+d)	2	10	C Z N	Push the 16-bit word at P2 + d onto the P3 stack, high byte first (little-endian at the new P3+1) -- the C compiler's argument push straight from a frame slot. A! (the address add); flags from that add.
\$BF	PHW (P3+d)	2	10	C Z N	Push the 16-bit word at P3 + d onto the P3 stack, high byte first (little-endian at the new P3+1) -- the C compiler's argument push straight from a frame slot; d is measured BEFORE the push. A! (the address add); flags from that add.
Tier A: the C-compiler ISA (2026-09; pure microcode. A! = clobbers A; d = unsigned 8-bit displacement)					
\$38	LDP1 #imm16	3	5	-	P1 := imm16. A real 3-byte instruction (was the LPL1/LPH1 pseudo-op pair).
\$39	LDP2 #imm16	3	5	-	P2 := imm16.
\$3A	LDP3 #imm16	3	5	-	P3 := imm16.
\$3C	ADDP3 #imm	2	6	C Z N	P3 := P3 + imm8 (free a stack frame). A!; flags are the low byte's.
\$3D	SUBP3 #imm	2	6	C Z N	P3 := P3 - imm8 (allocate a stack frame). A!; flags are the low byte's.
\$90	LDW addr,(P1+d)	4	14	C Z N	Word at addr := the word at P1 + d (a frame local into a memory word). A!

Op	Mnemonic	Bytes	Cycles	Flags	Description
\$91	LDW $\text{addr}, (\text{P2}+\text{d})\ 4$		14	C Z N	Word at $\text{addr} :=$ the word at $\text{P2} + \text{d}$ (a frame local into a memory word). A!
\$92	LDW $\text{addr}, (\text{P3}+\text{d})\ 4$		14	C Z N	Word at $\text{addr} :=$ the word at $\text{P3} + \text{d}$ (a frame local into a memory word). A!
\$94	STW $(\text{P1}+\text{d}), \text{addr}\ 4$		14	C Z N	Word at $\text{P1} + \text{d} :=$ the word at addr (a memory word into a frame local). A!
\$95	STW $(\text{P2}+\text{d}), \text{addr}\ 4$		14	C Z N	Word at $\text{P2} + \text{d} :=$ the word at addr (a memory word into a frame local). A!
\$96	STW $(\text{P3}+\text{d}), \text{addr}\ 4$		14	C Z N	Word at $\text{P3} + \text{d} :=$ the word at addr (a memory word into a frame local). A!
\$98	LDW $\text{addr}, \#imm8\ 4$	4	8	-	Word at $\text{addr} :=$ imm8 zero-extended (4 bytes; the compiler's constant idiom).
\$99	LDW $\text{addr}, \#imm16\ 5$	5	9	-	Word at $\text{addr} :=$ imm16 (5 bytes).
\$9A	ADDW $\text{dst}, \text{src}\ 5$	5	15	C Z N V	Word $a := a + b$ (16-bit, carry chained). C = carry out; N/V from the high byte; Z from the HIGH byte only. A!
\$9B	SUBW $\text{dst}, \text{src}\ 5$	5	15	C Z N V	Word $a := a - b$ (16-bit, borrow chained). C=1 means no borrow (unsigned $a \geq b$). Z high byte only. A!
\$9C	CMPW $\text{dst}, \text{src}\ 5$	5	15	C Z N V	Flags from $a - b$ (16-bit), memory unchanged: C = unsigned $a \geq b$; BLT/BGE/BLE/BGT give the signed order. Z high byte only. A!
\$9E	INCW $\text{addr}\ 3$	3	9	C Z N	Word at $\text{addr} :=$ word + 1. A!; flags are the low byte's (C = carry out of it).
\$9F	DECW $\text{addr}\ 3$	3	9	C Z N	Word at $\text{addr} :=$ word - 1. A!; flags are the low byte's (C=1: no borrow).
\$A0	ADDW $\text{addr}, \#imm8\ 4$	4	15	C Z N V	Word $a := a + imm8$ (zero-extended), 16-bit; C = carry out; Z of the FULL word. A!
\$A1	SUBW $\text{addr}, \#imm8\ 4$	4	15	C Z N V	Word $a := a - imm8$; C=1 means no borrow (unsigned $a \geq imm$); Z of the full word. A!
\$A2	CMPW $\text{addr}, \#imm8\ 4$	4	15	C Z N V	Flags from $a - imm8$ (16-bit), memory unchanged: C = unsigned $a \geq imm$; Z = ($a == imm$) over the full word; BLT/BGE signed. A!
\$A4	LEAW $\text{addr}, (\text{P1}+\text{d})\ 4$	4	14	C Z N	Word at $\text{addr} := \text{P1} + \text{d}$ -- the ADDRESS of a frame local (arrays, &x;). A!
\$A5	LEAW $\text{addr}, (\text{P2}+\text{d})\ 4$	4	14	C Z N	Word at $\text{addr} := \text{P2} + \text{d}$ -- the ADDRESS of a frame local (arrays, &x;). A!
\$A6	LEAW $\text{addr}, (\text{P3}+\text{d})\ 4$	4	14	C Z N	Word at $\text{addr} := \text{P3} + \text{d}$ -- the ADDRESS of a frame local (arrays, &x;). A!
\$B1	ADDW $\text{addr}, \#imm16\ 5$	5	15	C Z N V	Word $a := a + imm16$; C = carry out; Z of the full word. A! (5 bytes: `x + &table;`.)
\$B2	SUBW $\text{addr}, \#imm16\ 5$	5	15	C Z N V	Word $a := a - imm16$; C=1 means no borrow; Z of the full word. A!
\$B3	CMPW $\text{addr}, \#imm16\ 5$	5	15	C Z N V	Flags from $a - imm16$, memory unchanged: C = unsigned $a \geq imm$; Z = ($a == imm$); BLT/BGE/BLE/BGT signed. A!
\$B4	ANDW $\text{dst}, \text{src}\ 5$	5	15	Z N	Word $a := a \text{ AND } b$ (16-bit). Z from the high byte only. A!
\$B5	ORW $\text{dst}, \text{src}\ 5$	5	15	Z N	Word $a := a \text{ OR } b$ (16-bit). Z high byte only. A!
\$B6	XORW $\text{dst}, \text{src}\ 5$	5	15	Z N	Word $a := a \text{ XOR } b$ (16-bit). Z high byte only. A!
\$B7	ANDW $\text{addr}, \#imm8\ 4$	4	15	Z N	Word $a := a \text{ AND } imm8$ -- the high byte is ANDed with 0, i.e. cleared (a mask is a mask). Z of the full word, so `ANDW x, #1 ; JZ` tests a bit of a word. A!
\$B8	ORW $\text{addr}, \#imm8\ 4$	4	15	Z N	Word $a := a \text{ OR } imm8$ (high byte kept). Z of the full word. A!
\$B9	XORW $\text{addr}, \#imm8\ 4$	4	15	Z N	Word $a := a \text{ XOR } imm8$ (high byte kept). Z of the full word. A!
\$BA	ANDW $\text{addr}, \#imm16\ 5$	5	15	Z N	Word $a := a \text{ AND } imm16$. Z of the full word. A!
\$BB	ORW $\text{addr}, \#imm16\ 5$	5	15	Z N	Word $a := a \text{ OR } imm16$. Z of the full word. A!

Op	Mnemonic	Bytes	Cycles	Flags	Description
\$BC	XORW addr, #imm16	3	15	Z N	Word a := a XOR imm16; #\$FFFF is a 16-bit bitwise NOT (the compiler's ~x, and -x with INCW). Z of the full word. A!
Control flow					
\$40	JMP addr	3	4	-	P0 (PC) := addr.
\$41	JSR (P1)	1	9	-	Push return address (high byte first) onto P3 stack, then P0 := P1. Target must already be in P1.
\$42	RTS	1	6	-	Pop return address from P3 stack into P0.
\$43	JSR addr	3	10	-	Push return address, then P0 := addr (absolute call).
\$48	BZ addr	3	4	-	Branch to addr if Z=1.
\$49	BNZ addr	3	4	-	Branch to addr if Z=0.
\$4A	BCP addr	3	4	-	Branch if C=1, i.e. the RAW 74181 Cn+4 pin is high. Pin high means NO carry out - see note (2).
\$4C	JNC addr	3	4	-	Branch to addr if C=0. (JC/JZ/JNZ are aliases of BCP/BZ/BNZ.)
\$A8	JMP rel8	2	9	C Z N V	P0 := P0 + rel8 (signed, from the next instruction). 2 bytes; A! flags!; via .relax or JMP.R. An always-taken jump is cheaper as the 3-step absolute JMP (JMP.A), which the compiler emits.
\$A9	BZ rel8	2	9	C Z N V	Branch rel8 if Z=1. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes. (JZ.R alias.)
\$AA	BNZ rel8	2	9	C Z N V	Branch rel8 if Z=0. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes. (JNZ.R alias.)
\$AB	BCP rel8	2	9	C Z N V	Branch rel8 if C=1. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes. (JC.R alias.)
\$AC	JNC rel8	2	9	C Z N V	Branch rel8 if C=0. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes.
Signed branches (rev C; after CMP — N^V/Z)					
\$44	BLT addr	3	4	-	Branch if signed A < B (N^V=1). Use after CMP.
\$45	BGE addr	3	4	-	Branch if signed A >= B (N^V=0). Use after CMP.
\$46	BLE addr	3	4	-	Branch if signed A <= B ((N^V) Z). Use after CMP.
\$47	BGT addr	3	4	-	Branch if signed A > B (not (N^V) Z). Use after CMP.
\$AD	BLT rel8	2	9	C Z N V	Branch rel8 if signed A < B (N^V=1). Use after CMP. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes.
\$AE	BGE rel8	2	9	C Z N V	Branch rel8 if signed A >= B (N^V=0). Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes.
\$AF	BLE rel8	2	9	C Z N V	Branch rel8 if signed A <= B ((N^V) Z). Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes.
\$B0	BGT rel8	2	9	C Z N V	Branch rel8 if signed A > B. Taken: A and the flags are clobbered (8 steps); not taken: 2 steps, nothing changes.

Notes

(1) Shifts & rotates: SHL/SHR shift in 0 and latch the shifted-out bit into C. ROL/ROR rotate through C (the shifted-in bit is the current C). This makes multi-byte shifts work the conventional way (SHL low byte, then ROL high byte).

(2) C flag (rev B): CONVENTIONAL active-high carry. After ADD, C=1 means a carry occurred; after SUB/CMP, C=1 means no borrow (A >= B). JC/BCP branch on C=1, JNC on C=0. CLC/SEC clear/set C without disturbing Z/N/V. (Rev A latched the raw active-low 74181 Cn+4 pin; rev B embraces the conventional convention.)

(3) Stack: JSR pushes the return address high byte then low byte, decrementing after each write (empty-descending). RTS increments then reads. Software must initialise P3 (e.g. LDP3 #\$FEFF) before the first JSR. **(4) V flag:** reads 0 in rev A. **(5) Absolute addressing:** LDA/LDB/STA/JSR accept an absolute address; the hardware forms it in the hidden PT scratch pointer.

Assembler quick reference (p8xasm.py)

```

label:  MNEMONIC operand          ; comment
operands:  #expr (immediate) | (Pn) / (Pn)+ | expr (16-bit address)
exprs:    $1F 0x1F 31 'c' symbol with +/-, <expr = low byte, >expr = high
directives: .org e .byte e,... .word e,... .ascii "s" .asciiz "s"
            .fill n[,v] NAME = expr .equ NAME, expr
pseudo-op: LDPn #imm16           ; expands to LPLn #<imm, LPHn #>imm
usage:     python3 p8xasm.py prog.asm -o eeeprom.bin [-l prog.lst]

```

Example: print a string

```

ACIA_D = $FF05
        .org 0
        LDP2 #ACIA_D           ; P2 -> ACIA data register
        LDP1 #msg              ; P1 -> string
        LDB #0
loop:    LDA (P1)+              ; fetch byte, advance
        OR                     ; A := A|0 - sets Z on the terminator
        BZ done
        STA (P2)               ; transmit
        JMP loop
done:    HLT
msg:     .asciiz "P8X lives!\r\n"

```

Writing a program for P8X/OS

Programs launched by the OS **RUN** command load into the transient program area at **\$5900** and run via a **JSR** to their exec address. The program ABI: **return to the shell with RTS** (P3, the stack, is the OS's - leave it balanced); on entry **P2 points at the argument tail** - the command text after the program name, NUL-terminated (so `RUN EDIT F00.ASM` enters with P2 -> "F00.ASM"); programs that take no arguments ignore P2. Build with `.org $5900` and the host assembler's `--base 0x5900`, or assemble on-target with ASM. A file created on-target carries load/exec 0, which the OS maps to \$5900, so it is directly RUNnable. The BIOS jump table at \$0100 (console + CF, the FFIND/FCREATE/FDELETE/FCOMMIT file calls, the FOPEN/FGETB and FWOPEN/FPUTB/FCLOSE byte streams, and FRESOLVE/FNORM/FOPENDIR/FNEXT) is the only entry point a program needs - it must not call into OS internals.